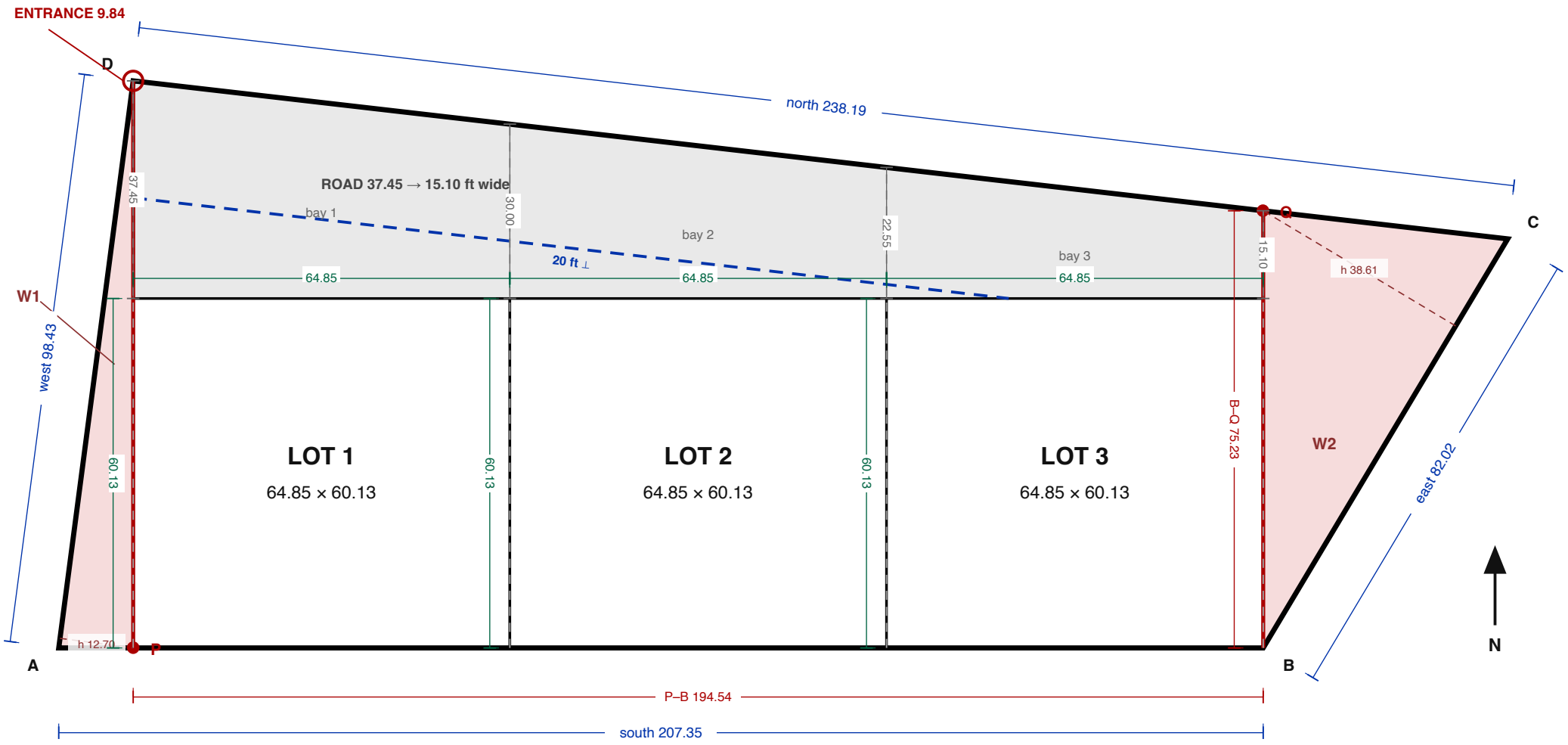
**TS 11/14/18 · Ward-16 Thallakulam · Madurai North**

OPTION A — LEVEL

scale 1:300 — print at 100% on A4

north up · all figures in feet




**SCALE 1:300 — 1 mm on paper = 0.30 m = 0.98 ft on the ground**

0 5 10 15 20 25 m

this bar is exactly 100 mm — check it with a ruler; if it differs the print was resized

## TS 11/14/18 — OPTION A — LEVEL — SCHEDULE

lots 61.5% · road 26.9% · wastage 11.6%

perpendicular down from D → P · perpendicular up from B → Q · three identical rectangles · road 15.00 ft ⊥ at its narrowest, opening to 37.45 ft

sides 72.60 / 25.00 / 63.20 / 30.00 m · closure = 1766.9 m² = 19,019 ft² = 43.66 cents · W1 = west corner triangle A-P-D, W2 = east corner triangle B-Q-C

closure: Heron(63.20, 25.00, 79.00) + Heron(79.00, 72.60, 30.00) = 678.0 + 1088.9 = 1766.9 m²

corner angles A 82.52° B 120.88° C 65.68° D 90.92° · A+C = 148.20° so the quadrilateral is NOT cyclic — Brahmagupta's 1838.5 m² is out of reach for this shape · diagonals AC 79.00 m, BD 66.34 m

## SCHEDULE — every side length in feet, every area from its own arithmetic

| piece | sides (ft) | area = base × height | area ft² | cents | m² |
|---|---|---|---|---|---|
| **Lot 1** | **64.85 × 60.13 × 64.85 × 60.13** | **64.85 × 60.13** | **3,899** | **8.95** | **362.3** |
| **Lot 2** | **64.85 × 60.13 × 64.85 × 60.13** | **64.85 × 60.13** | **3,899** | **8.95** | **362.3** |
| **Lot 3** | **64.85 × 60.13 × 64.85 × 60.13** | **64.85 × 60.13** | **3,899** | **8.95** | **362.3** |
| road bay 1 (over Lot 1) | 64.85 × 30.00 × 65.27 × 37.45 | (37.45 + 30.00)/2 × 64.85 | 2,187 | 5.02 | 203.2 |
| road bay 2 (over Lot 2) | 64.85 × 22.55 × 65.27 × 30.00 | (30.00 + 22.55)/2 × 64.85 | 1,704 | 3.91 | 158.3 |
| road bay 3 (over Lot 3) | 64.85 × 15.10 × 65.27 × 22.55 | (22.55 + 15.10)/2 × 64.85 | 1,221 | 2.80 | 113.4 |
| **LOTS + ROAD** | | | **16,810** | **38.59** | **1561.7** |
| W1 (A-P-D) | 12.81 × 97.59 × 98.43 | ½ × 98.43 × 12.70 | 625 | 1.44 | 58.1 |
| W2 (B-Q-C) | 75.23 × 42.37 × 82.02 | ½ × 82.02 × 38.61 | 1,583 | 3.64 | 147.1 |
| **WASTAGE** | | | **2,209** | **5.07** | **205.2** |
| **TOTAL** | | | **19,019** | **43.66** | **1766.9** |

**TALLY 16,810 (lots + road) + 2,209 (wastage) = 19,019 ft² = 43.66 cents = 1766.9 m² = the whole plot**

REGISTERED EXTENT is 1875 m² (20,182 ft² / 46.33 cents) — 108.1 m² more than these four sides can close (absolute maximum 1838.5 m²).

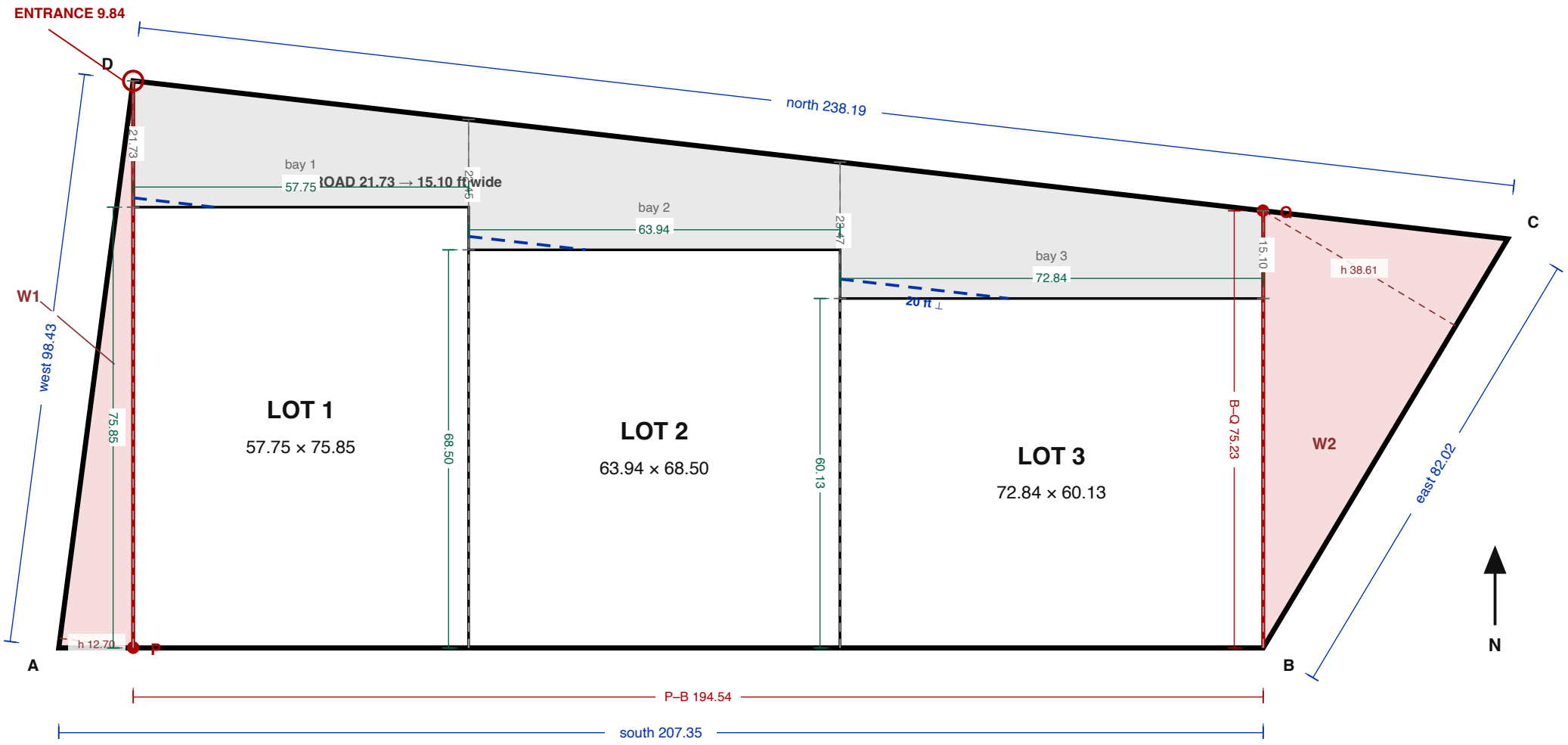
To price on the register multiply every area above by 1.0612: lot = 4,138 ft² = 9.50 cents. The fitted-sides option does this by stretching the boundary instead.

**TS 11/14/18 · Ward-16 Thallakulam · Madurai North**

OPTION B — STEPPED

scale 1:300 — print at 100% on A4

north up · all figures in feet




**SCALE 1:300 — 1 mm on paper = 0.30 m = 0.98 ft on the ground**

0 5 10 15 20 25 m

this bar is exactly 100 mm — check it with a ruler; if it differs the print was resized

## TS 11/14/18 — OPTION B — STEPPED — SCHEDULE

lots 69.1% · road 19.3% · wastage 11.6%

perpendicular down from D → P · perpendicular up from B → Q · rear line steps down west→east · road 15.00 ft ⊥ at its narrowest, opening to 21.73 ft

sides 72.60 / 25.00 / 63.20 / 30.00 m · closure = 1766.9 m² = 19,019 ft² = 43.66 cents · W1 = west corner triangle A-P-D, W2 = east corner triangle B-Q-C

closure: Heron(63.20, 25.00, 79.00) + Heron(79.00, 72.60, 30.00) = 678.0 + 1088.9 = 1766.9 m²

corner angles A 82.52° B 120.88° C 65.68° D 90.92° · A+C = 148.20° so the quadrilateral is NOT cyclic — Brahmagupta's 1838.5 m² is out of reach for this shape · diagonals AC 79.00 m, BD 66.34 m

## SCHEDULE — every side length in feet, every area from its own arithmetic

| piece | sides (ft) | area = base × height | area ft² | cents | m² |
|---|---|---|---|---|---|
| **Lot 1** | **57.75 × 75.85 × 57.75 × 75.85** | **57.75 × 75.85** | **4,380** | **10.06** | **407.0** |
| **Lot 2** | **63.94 × 68.50 × 63.94 × 68.50** | **63.94 × 68.50** | **4,380** | **10.06** | **407.0** |
| **Lot 3** | **72.84 × 60.13 × 72.84 × 60.13** | **72.84 × 60.13** | **4,380** | **10.06** | **407.0** |
| road bay 1 (over Lot 1) | 57.75 × 15.10 × 58.13 × 21.73 | (21.73 + 15.10)/2 × 57.75 | 1,064 | 2.44 | 98.8 |
| road bay 2 (over Lot 2) | 63.94 × 15.10 × 64.36 × 22.45 | (22.45 + 15.10)/2 × 63.94 | 1,200 | 2.76 | 111.5 |
| road bay 3 (over Lot 3) | 72.84 × 15.10 × 73.32 × 23.47 | (23.47 + 15.10)/2 × 72.84 | 1,405 | 3.22 | 130.5 |
| **LOTS + ROAD** | | | **16,810** | **38.59** | **1561.7** |
| W1 (A-P-D) | 12.81 × 97.59 × 98.43 | ½ × 98.43 × 12.70 | 625 | 1.44 | 58.1 |
| W2 (B-Q-C) | 75.23 × 42.37 × 82.02 | ½ × 82.02 × 38.61 | 1,583 | 3.64 | 147.1 |
| **WASTAGE** | | | **2,209** | **5.07** | **205.2** |
| **TOTAL** | | | **19,019** | **43.66** | **1766.9** |

**TALLY 16,810 (lots + road) + 2,209 (wastage) = 19,019 ft² = 43.66 cents = 1766.9 m² = the whole plot**

REGISTERED EXTENT is 1875 m² (20,182 ft² / 46.33 cents) — 108.1 m² more than these four sides can close (absolute maximum 1838.5 m²).

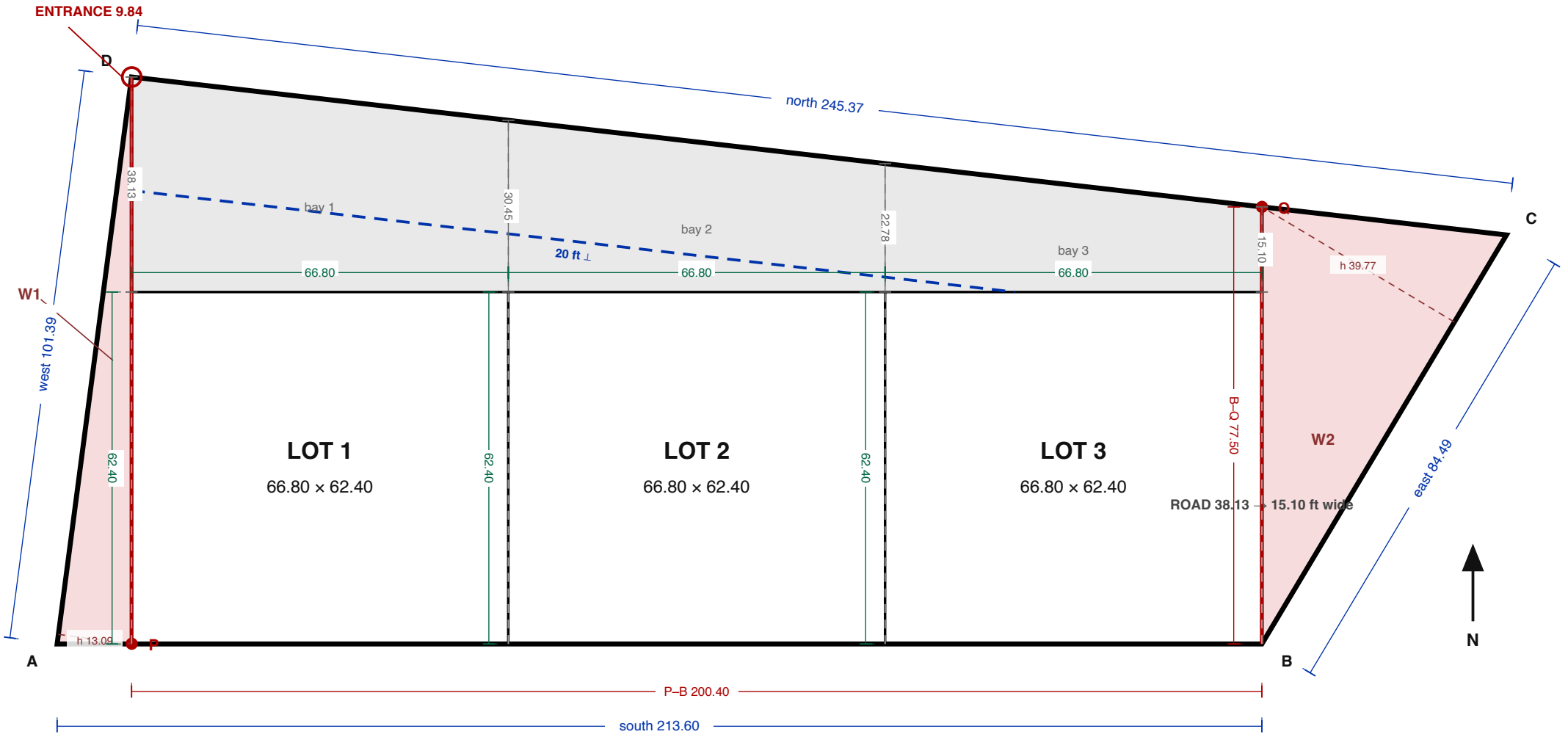
To price on the register multiply every area above by 1.0612: lot = 4,648 ft² = 10.67 cents. The fitted-sides option does this by stretching the boundary instead.

**TS 11/14/18 · Ward-16 Thallakulam · Madurai North**

OPTION C — LEVEL, SIDES FITTED TO 1875 m²

scale 1:300 — print at 100% on A4

north up · all figures in feet




**SCALE 1:300 — 1 mm on paper = 0.30 m = 0.98 ft on the ground**

0 5 10 15 20 25 m

this bar is exactly 100 mm — check it with a ruler; if it differs the print was resized

## TS 11/14/18 — OPTION C — LEVEL, SIDES FITTED TO 1875 m² — SCHEDULE

lots 62.0% · road 26.4% · wastage 11.6%

perpendicular down from D → P · perpendicular up from B → Q · three identical rectangles · road 15.00 ft ⊥ at its narrowest, opening to 38.13 ft

sides 74.79 / 25.75 / 65.10 / 30.90 m · closure = 1875.0 m² = 20,182 ft² = 46.33 cents · W1 = west corner triangle A-P-D, W2 = east corner triangle B-Q-C

closure: Heron(65.10, 25.75, 81.38) + Heron(81.38, 74.79, 30.90) = 719.5 + 1155.5 = 1875.0 m²

corner angles A 82.52° B 120.88° C 65.68° D 90.92° · A+C = 148.20° so the quadrilateral is NOT cyclic — Brahmagupta's 1951.0 m² is out of reach for this shape · diagonals AC 81.38 m, BD 68.34 m

## SCHEDULE — every side length in feet, every area from its own arithmetic

| piece | sides (ft) | area = base × height | area ft² | cents | m² |
|---|---|---|---|---|---|
| **Lot 1** | **66.80 × 62.40 × 66.80 × 62.40** | **66.80 × 62.40** | **4,168** | **9.57** | **387.3** |
| **Lot 2** | **66.80 × 62.40 × 66.80 × 62.40** | **66.80 × 62.40** | **4,168** | **9.57** | **387.3** |
| **Lot 3** | **66.80 × 62.40 × 66.80 × 62.40** | **66.80 × 62.40** | **4,168** | **9.57** | **387.3** |
| road bay 1 (over Lot 1) | 66.80 × 30.45 × 67.24 × 38.13 | (38.13 + 30.45)/2 × 66.80 | 2,291 | 5.26 | 212.8 |
| road bay 2 (over Lot 2) | 66.80 × 22.78 × 67.24 × 30.45 | (30.45 + 22.78)/2 × 66.80 | 1,778 | 4.08 | 165.2 |
| road bay 3 (over Lot 3) | 66.80 × 15.10 × 67.24 × 22.78 | (22.78 + 15.10)/2 × 66.80 | 1,265 | 2.90 | 117.5 |
| **LOTS + ROAD** | | | **17,839** | **40.95** | **1657.3** |
| W1 (A-P-D) | 13.20 × 100.53 × 101.39 | ½ × 101.39 × 13.09 | 663 | 1.52 | 61.6 |
| W2 (B-Q-C) | 77.50 × 43.65 × 84.49 | ½ × 84.49 × 39.77 | 1,680 | 3.86 | 156.1 |
| **WASTAGE** | | | **2,344** | **5.38** | **217.7** |
| **TOTAL** | | | **20,182** | **46.33** | **1875.0** |

**TALLY 17,839 (lots + road) + 2,344 (wastage) = 20,182 ft² = 46.33 cents = 1875.0 m² = the whole plot**

SIDES STRETCHED 3.01% from the FMB's 72.60 / 25.00 / 63.20 / 30.00 m so the closure equals the REGISTERED 1875 m² exactly.

Shape and every angle unchanged: each length ×1.03014, each area ×1.0612. Verify on the ground before relying on it.